ALGEBRA 1 · PERSONALIZED PRACTICE

Math Practice

2026-11-13 · 20 questions

I CAN USE INPUT-OUTPUT RULES TO REPRESENT RELATIONSHIPS.

1. I can use input-output rules to represent relationships.

Rules A and B are $2x + 1$ and $x + 3$. At which input do they give the same output?

2. I can use input-output rules to represent relationships.

Input	-1	0	2
Output	1	3	7

Describe an operation rule that works for every pair.

3. I can use input-output rules to represent relationships.

A machine adds 4 to an input, then squares the result. Explain why $(x + 4)^2$ matches but $x^2 + 4$ does not.

4. I can use input-output rules to represent relationships.

Write a rule.

Inputs 0,1,2 give outputs 4,7,10.

I CAN DISTINGUISH FUNCTIONS FROM RELATIONSHIPS THAT ARE NOT FUNCTIONS.

5. I can distinguish functions from relationships that are not functions.

Does $\{(1, 2), (1, 3), (2, 4)\}$ represent a function? Explain.

6. I can distinguish functions from relationships that are not functions.

Is $(1, 3), (2, 3), (3, 3)$ a function? Evaluate $2(-3) + 5$.

7. I can distinguish functions from relationships that are not functions.

Pairs $(1, 5), (2, 5), (3, 5)$ all have output 5. Is this a function? Explain.

8. I can distinguish functions from relationships that are not functions.

Identify the repeated input.

x	-1	2	-1
y	3	5	8

I CAN IDENTIFY THE DOMAIN AND RANGE OF A RELATION.

9. I can identify the domain and range of a relation.

The function $f(x) = \sqrt{x}$ has real outputs. Explain why negative inputs are excluded.

10. I can identify the domain and range of a relation.

Tickets cost \$6 each, and at most 8 tickets may be bought. What inputs are reasonable for the total-cost rule?

11. I can identify the domain and range of a relation.

List the possible inputs.

A parking garage has levels 1 through 5. List the possible level inputs.

12. I can identify the domain and range of a relation.

List the domain.

$(2, 5), (4, 7), (6, 5)$

I CAN CONNECT LINEAR GROWTH IN TABLES, GRAPHS, AND EQUATIONS.

13. I can connect linear growth in tables, graphs, and equations.

Find the slope.

Points (1,3) and (5,11)

14. I can connect linear growth in tables, graphs, and equations.

Find the change per step for outputs 5, 9, 13, 17. Is $(1, 2), (2, 2), (1, 3)$ a function?

15. I can connect linear growth in tables, graphs, and equations.

The outputs 2, 5, 8, 11 increase by a constant amount. What is the change per step?

16. I can connect linear growth in tables, graphs, and equations.

Complete the table.

x	0	1	2	3
y	4	7	10	?

I CAN CALCULATE AND INTERPRET SLOPE AS A CONSTANT RATE OF CHANGE.

17. I can calculate and interpret slope as a constant rate of change.

As x rises by 4, y falls by 12. Find the slope.

18. I can calculate and interpret slope as a constant rate of change.

Find the rise.

(3,−4) to (7,5)

19. I can calculate and interpret slope as a constant rate of change.

Find the slope.

$(-2,1), (2,9)$

20. I can calculate and interpret slope as a constant rate of change.

Find the slope through $(0, 2)$ and $(3, 8)$.